



Habib University
shaping futures

Instructions

- The exam will begin at **5:40 PM** and conclude at **7:10 PM**, with a total duration of **1 hour and 30 minutes**. The exam is worth **30 points** in total.
- This is a closed books/notes exam. Any notes, books, bags should be kept at the front of the classroom before the start of the test.
- Use of cellphones, laptops or any other communicating device is prohibited. Your cellphones should be switched off and submitted to the invigilator, or kept at the front of the class room before the start of the test.
- You are to attempt all question on the answer sheet provided. Your solutions should be comprehensible. Explain what you are doing, and if you use any results that you have studied in this course or Calculus 1 and 2, mention them explicitly.
- Use of calculators is allowed for this exam. Any form of communication/sharing among your peers is not allowed during the exam.
- If you want to communicate with the invigilator, raise your hand.
- You are not allowed to leave your desk without permission of the invigilator during the exam.
- Failure to abide by the instructions above would lead to immediate cancellation of the exam.

Student Name: _____

Instructor's Name: _____

Section: _____

1. (5 points) Show that for a planar region R with positively oriented, piecewise-smooth boundary $C = \partial R$,

$$\text{Area}(R) = \iint_R dx dy = -\frac{1}{2} \oint_C (y dx - x dy).$$

Here C is taken with the positive (counterclockwise) orientation.

Solution:

Proof using Green's Theorem.

By Green's Theorem (Circulation-Curl Form), for a positively oriented, piecewise-smooth, simple closed curve $C = \partial R$,

$$\oint_C (P dx + Q dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy.$$

We choose

$$P = -\frac{y}{2}, \quad Q = \frac{x}{2}.$$

Then

$$\frac{\partial Q}{\partial x} = \frac{1}{2}, \quad \frac{\partial P}{\partial y} = -\frac{1}{2}.$$

Hence,

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{1}{2} - \left(-\frac{1}{2}\right) = 1.$$

Substituting into Green's Theorem gives

$$\oint_C (P dx + Q dy) = \iint_R 1 dx dy = \text{Area}(R).$$

Finally, since

$$P dx + Q dy = -\frac{y}{2} dx + \frac{x}{2} dy = \frac{1}{2}(x dy - y dx),$$

we obtain

$$\boxed{\text{Area}(R) = \frac{1}{2} \oint_C (x dy - y dx).}$$

2. (7 points) Compute the flux of $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$ through the curved surface of the cylinder $x^2 + y^2 = 9$ bounded below by the plane $x + y + z = 2$, above by the plane $x + y + z = 5$, and oriented away from the z -axis.

Solution:

Given:

$$\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}, \quad x^2 + y^2 = 9,$$

bounded below by the plane $x + y + z = 2$ and above by $x + y + z = 5$, with the surface oriented *away from the z-axis*.

Step 1: Normal vector

For the cylinder $x^2 + y^2 = 9$, Consider $g = x^2 + y^2 - 9$

$$\vec{\nabla}g = 2x\vec{i} + 2y\vec{j}.$$

Hence, the outward unit normal vector is

$$\vec{n} = \frac{\vec{\nabla}g}{|\vec{\nabla}g|} = \frac{x\vec{i} + y\vec{j}}{4}.$$

Step 2: Flux formula

The flux through the curved surface is

$$\Phi = \iint_S \vec{F} \cdot \vec{n} \, d\sigma.$$

We parameterize the cylinder as

$$x = 3 \cos \theta, \quad y = 3 \sin \theta, \quad 0 \leq \theta \leq 2\pi.$$

For each (x, y) , the planes give

$$z_{\text{bottom}} = 2 - (x + y), \quad z_{\text{top}} = 5 - (x + y).$$

Step 3: Dot product $\vec{F} \cdot \vec{n}$

At any point on the cylinder,

$$\vec{F} \cdot \vec{n} = (x\vec{i} + y\vec{j} + z\vec{k}) \cdot \frac{x\vec{i} + y\vec{j}}{4} = \frac{1}{4}(x^2 + y^2) = \frac{9}{4}$$

Step 4: Surface element

For a cylinder of radius 4,

$$d\sigma = |\vec{n}| \, d\theta \, dz = 4 \, d\theta \, dz.$$

Step 5: Height of the cylindrical strip

At a fixed θ ,

$$z_{\text{top}} - z_{\text{bottom}} = [5 - (3 \cos \theta + 3 \sin \theta)] - [2 - (3 \cos \theta + 3 \sin \theta)] = 3.$$

Step 6: Compute flux

$$\Phi = \int_0^{2\pi} \int_{z_{\text{bottom}}}^{z_{\text{top}}} (\vec{F} \cdot \vec{n}) d\sigma = \int_0^{2\pi} \int_{z_{\text{bottom}}}^{z_{\text{top}}} \frac{9}{4}(4) dz d\theta = 9 \int_0^{2\pi} (z_{\text{top}} - z_{\text{bottom}}) d\theta.$$

Substituting $z_{\text{top}} - z_{\text{bottom}} = 3$,

$$\Phi = 9 \int_0^{2\pi} 3 d\theta = 27 \int_0^{2\pi} d\theta = 27(2\pi) = 54\pi.$$

$$\boxed{\Phi = 54\pi.}$$

3. (8 points) The region W lies between the spheres $x^2 + y^2 + z^2 = 4$ and $x^2 + y^2 + z^2 = 25$ and within the cone $z = \sqrt{x^2 + y^2}$ with $z \geq 0$; its boundary is the closed surface, S , oriented outward. Find the flux of $\vec{F} = x^3\vec{i} + y^3\vec{j} + z^3\vec{k}$ out of S .

Solution:

Given:

$$\vec{F} = x^3\vec{i} + y^3\vec{j} + z^3\vec{k}.$$

The region W lies between the spheres $x^2 + y^2 + z^2 = 4$ and $x^2 + y^2 + z^2 = 9$, and within the cone $z = \sqrt{x^2 + y^2}$ with $z \geq 0$. Let S be the closed surface of W , oriented outward.

Using the Divergence Theorem:

$$\iint_S \vec{F} \cdot \vec{n} d\sigma = \iiint_W (\vec{\nabla} \cdot \vec{F}) dV.$$

Compute the divergence:

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial x^3}{\partial x} + \frac{\partial y^3}{\partial y} + \frac{\partial z^3}{\partial z} = 3(x^2 + y^2 + z^2) = 3\rho^2.$$

In spherical coordinates:

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

The limits are:

$$2 \leq \rho \leq 5, \quad 0 \leq \phi \leq \frac{\pi}{4}, \quad 0 \leq \theta \leq 2\pi.$$

Hence,

$$\begin{aligned} \Phi &= \iiint_W 3\rho^2 \, dV = 3 \int_0^{2\pi} \int_0^{\pi/4} \int_2^5 \rho^2 (\rho^2 \sin \phi) \, d\rho \, d\phi \, d\theta \\ &= 3 \int_0^{2\pi} d\theta \int_0^{\pi/4} \sin \phi \, d\phi \int_2^5 \rho^4 \, d\rho. \end{aligned}$$

Evaluate:

$$\begin{aligned} \int_2^5 r^4 \, dr &= \frac{1}{5}(5^5 - 2^5) = \frac{3093}{5}, \\ \int_0^{\pi/4} \sin \theta \, d\theta &= 1 - \cos \frac{\pi}{4} = 1 - \frac{\sqrt{2}}{2}, \quad \int_0^{2\pi} d\phi = 2\pi. \end{aligned}$$

Therefore,

$$\Phi = 3 \times 2\pi \times \left(1 - \frac{\sqrt{2}}{2}\right) \times \frac{3093}{5} = \frac{18558\pi}{5} \left(1 - \frac{\sqrt{2}}{2}\right).$$

$$\boxed{\Phi = \frac{18558\pi}{5} \left(1 - \frac{\sqrt{2}}{2}\right)}.$$

4. (10 points) Suppose

$$\vec{F}(x, y, z) = -z^2 \vec{j} + yz \vec{k}$$

1. Does there exist a vector field $\vec{G}$ such that $\vec{\nabla} \times \vec{G} = \vec{F}$? Explain why or why not.
2. Is $\vec{F}$ a gradient vector field? Justify your answer.
3. Let $S = \{(x, y, z) : x^2 + y^2 + z^2 = 4, x \geq 0\}$. Compute the surface integral

$$\iint_S \vec{\nabla} \times \vec{F} \cdot \vec{n} \, d\sigma$$

without using any integral theorems by choosing a convenient parametrization of S .

4. Verify your answer (3) using Stokes' Theorem and evaluating the appropriate integral.

Solution:

Given:

$$\vec{F}(x, y, z) = 0\vec{i} - z^2\vec{j} + yz\vec{k}.$$

1. **Existence of $\vec{G}$ such that $\vec{\nabla} \times \vec{G} = \vec{F}$**

A necessary condition for a vector field $\vec{F}$ to be a curl of some vector field $\vec{G}$ is that

$$\vec{\nabla} \cdot \vec{F} = 0,$$

since $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{G}) \equiv 0$ for any $\vec{G}$.

Compute:

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(-z^2) + \frac{\partial}{\partial z}(yz) = 0 + 0 + y = y.$$

Since $\vec{\nabla} \cdot \vec{F} = y \neq 0$ in general, it follows that there is **no** vector field $\vec{G}$ satisfying $\vec{\nabla} \times \vec{G} = \vec{F}$ on $\mathbb{R}^3$.

No such $\vec{G}$ exists.

2. **Is $\vec{F}$ a gradient field?**

A necessary and sufficient condition (on a simply connected region) for $\vec{F}$ to be a gradient field is that its curl vanish:

$$\vec{\nabla} \times \vec{F} = \vec{0}.$$

Compute:

$$\begin{aligned} \vec{\nabla} \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial_x & \partial_y & \partial_z \\ 0 & -z^2 & yz \end{vmatrix} \\ &= \left(\frac{\partial}{\partial y}(yz) - \frac{\partial}{\partial z}(-z^2) \right) \vec{i} - \left(\frac{\partial}{\partial x}(yz) - \frac{\partial}{\partial z}(0) \right) \vec{j} + \left(\frac{\partial}{\partial x}(-z^2) - \frac{\partial}{\partial y}(0) \right) \vec{k}. \end{aligned}$$

Simplify each term:

$$\frac{\partial}{\partial y}(yz) = z, \quad \frac{\partial}{\partial z}(-z^2) = -2z, \quad \frac{\partial}{\partial x}(yz) = 0, \quad \frac{\partial}{\partial x}(-z^2) = 0.$$

Thus,

$$\vec{\nabla} \times \vec{F} = (z - (-2z))\vec{i} + 0\vec{j} + 0\vec{k} = 3z\vec{i}.$$

Since $\vec{\nabla} \times \vec{F} = 3z\vec{i} \neq \vec{0}$ in general, $\vec{F}$ is **not** a gradient vector field.

$\vec{F}$ is not conservative.

3. **Compute** $\iint_S (\vec{\nabla} \times \vec{F}) \cdot \vec{n} \, d\sigma$ **directly, for** $S = \{(x, y, z) : x^2 + y^2 + z^2 = 4, x \geq 0\}$.

From (2),

$$\vec{\nabla} \times \vec{F} = 3z\vec{i}.$$

On the unit sphere, the outward unit normal is $\vec{n} = (x, y, z)$, so

$$(\vec{\nabla} \times \vec{F}) \cdot \vec{n} = (3z, 0, 0) \cdot (x, y, z) = 3xz.$$

Using spherical coordinates:

$$x = 2 \sin \phi \cos \theta, \quad y = 2 \sin \phi \sin \theta, \quad z = 2 \cos \phi,$$

with $0 \leq \phi \leq \pi$ and $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ (since $x \geq 0$).

The surface element is $d\sigma = 4 \sin \phi \, d\phi \, d\theta$. Hence,

$$\iint_S (\vec{\nabla} \times \vec{F}) \cdot \vec{n} \, d\sigma = \int_{-\pi/2}^{\pi/2} \int_0^\pi 3(2 \sin \phi \cos \theta)(2 \cos \phi) 4 \sin \phi \, d\phi \, d\theta.$$

Simplify:

$$= 48x \int_{-\pi/2}^{\pi/2} \cos \theta \, d\theta \int_0^\pi \sin^2 \phi \cos \phi \, d\phi.$$

Let $u = \sin \phi$, so $du = \cos \phi \, d\phi$:

$$\int_0^\pi \sin^2 \phi \cos \phi \, d\phi = \int_0^0 u^2 \, du = 0.$$

Therefore,

$$\iint_S (\vec{\nabla} \times \vec{F}) \cdot \vec{n} \, d\sigma = 0.$$

4. Verification using Stokes' Theorem

By Stokes' Theorem,

$$\iint_S (\vec{\nabla} \times \vec{F}) \cdot \vec{n} \, d\sigma = \oint_{\partial S} \vec{F} \cdot d\vec{r}.$$

The boundary ∂S is the circle where $x = 0$ and $y^2 + z^2 = 4$.

Parametrize it by

$$\vec{r}(t) = (0, 2 \cos t, 2 \sin t), \quad 0 \leq t \leq 2\pi.$$

Then $d\vec{r} = (0, -2 \sin t, 2 \cos t) \, dt$.

On this circle,

$$\vec{F} = -z^2 \vec{j} + yz \vec{k} = -4 \sin^2 t \vec{j} + (4 \cos t \sin t) \vec{k}.$$

Hence,

$$\begin{aligned} \vec{F} \cdot d\vec{r} &= (-4 \sin^2 t)(-2 \sin t) \, dt + (4 \cos t \sin t)(2 \cos t) \, dt \\ &= 8 \sin^3 t \, dt + 8 \cos^2 t \sin t \, dt \\ &= 8 \sin t (\sin^2 t + \cos^2 t) \, dt = 8 \sin t \, dt. \end{aligned}$$

Thus,

$$\oint_{\partial S} \vec{F} \cdot d\vec{r} = 8 \int_0^{2\pi} \sin t \, dt = 0.$$

Therefore, the result agrees with the direct computation.

$$\boxed{\iint_S (\vec{\nabla} \times \vec{F}) \cdot \vec{n} \, d\sigma = \oint_{\partial S} \vec{F} \cdot d\vec{r} = 0.}$$

Concept	Formula
Green's Theorem (Circulation-Curl Form)	$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$
Green's Theorem (Flux-Divergence Form)	$\oint_C P dy - Q dx = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA$
Surface Element $d\vec{\sigma}$	Parametric: $d\vec{\sigma} = (\vec{r}_u \times \vec{r}_v) dudv$ Implicit: $d\vec{\sigma} = \frac{\vec{\nabla} F}{ \vec{\nabla} F \cdot \vec{p} } dA$ Graph $z = f(x, y)$: $d\vec{\sigma} = \langle -f_x, -f_y, 1 \rangle dxdy$
Surface Area (Parametric)	$\text{Area} = \iint_D \vec{r}_u \times \vec{r}_v dudv$
Surface Area (Implicit Surface $F(x, y, z) = 0$)	$\text{Area}(S) = \iint_D \frac{ \vec{\nabla} F(x, y, z) }{ \vec{\nabla} F(x, y, z) \cdot \vec{p} } dA$ where $p = \vec{i}, \vec{j}$, or $\vec{k}$ is normal to R and $\nabla F \cdot \vec{p} \neq 0$.
Surface Area (Graph $z = f(x, y)$)	$\text{Area} = \iint_D \sqrt{1 + f_x^2 + f_y^2} dxdy$
Scalar Surface Integral	$\iint_S f(x, y, z) d\sigma = \iint_D f(\vec{r}(u, v)) \vec{r}_u \times \vec{r}_v dudv$
Vector Surface Integral (Flux)	$\iint_S \vec{F} \cdot \vec{n} d\sigma$
Parametric Surface $\vec{r}(u, v)$	$\iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) dudv$
Implicit Surface $F(x, y, z) = 0$	$\iint_D \vec{F} \cdot \frac{\vec{\nabla} F}{ \vec{\nabla} F \cdot \vec{p} } d\sigma$
Graph $z = f(x, y)$	$\iint_D \vec{F} \cdot \langle -f_x, -f_y, 1 \rangle dxdy$
Divergence of a vector field $\vec{F} = \langle P, Q, R \rangle$	$\vec{\nabla} \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$
Curl of a vector field $\vec{F} = \langle P, Q, R \rangle$	$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{vmatrix} = \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle$
Divergence Theorem	$\iint_S \vec{F} \cdot \vec{n} d\sigma = \iiint_V \vec{\nabla} \cdot \vec{F} dV$
Stokes' Theorem	$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\vec{\nabla} \times \vec{F}) \cdot \vec{n} d\sigma$
Cylindrical Coordinates	$x = r \cos \theta, y = r \sin \theta, z = z, dV = r dr d\theta dz$
Spherical Coordinates	$x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi, dV = \rho^2 \sin \phi d\rho d\phi d\theta$